Buyu Deng

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BuyuDeng

Education

2023.09 – Present **Undergraduate**, *School of the Gifted Young*, University of Science and Technology of China (USTC)
Major: Geophysics

GPA: 3.61/4.30 (**Rank**: 3/20)
Relevant Coursework: Linear Algebra: **90**, Probability & Statistics: **92**, Theoretical Mechanics: **95**, Python & Deep Learning: **A**, Programming in Geophysics: **94**, Introduction to Seismic Exploration: **98**
Test Scores: TOEFL: **5.0/6.0** (R: 4.5, L: 5.5, S: 4.5, W: 5.5); GRE: V: **146**, Q: **165**

Research Experience

2024.07 – Present **Undergraduate Researcher**, *Computational Interpretation Group*
Supervisor: **Prof. Xinming Wu**

3D "All-in-One" Seismic Denoising

- Built a synthetic 3D seismic dataset with random noise, acquisition footprints, and migration artifacts; trained an all-in-one denoising network based on Degradation Representation Learning.
  Code: github.com/BuyuDeng/NDR_Seismic

Geo-VAE for 3D Geophysical Data Compression & Latent Processing

- Adapted Wan-VAE into a unified Geo-VAE for seismic volumes, RGT cubes, property models, structures, and geobodies, achieving 8×8×8 volumetric compression with high-fidelity reconstruction.
- Enabled memory-efficient layer-by-layer inference on large 3D volumes using causal 3D convolutions and feature caching, avoiding chunk-boundary artifacts.
- Built latent-space downstream workflows for missing-trace interpolation with a dual-mask latent diffusion model and seismic denoising with an HRNet-style latent mapper.
  Submission: full paper submitted to *Geophysics*; abstract selected as **Oral** at IMAGE 2026.
  Note: *Geophysics* special section "Machine Learning in Seismic Processing for a Sustainable Future".
  Code: github.com/BuyuDeng/Geo-VAE

PHY-to-BGC Ocean Biogeochemical Emulation

- Built a 2.5D full-image U-Net emulator mapping global physical ocean fields to coupled BGC variables. The model uses monthly anomaly learning, seasonal/latitudinal auxiliary channels, longitude-circular padding, and global/vertical/basin-scale evaluation.
- Demonstrated that a model trained on normal years maintains strong performance in an El Nino year, suggesting robust generalization under anomalous ocean-climate conditions; **manuscript in preparation**.

GNSS-NISAR Ionospheric Super-Resolution Reconstruction

- Developed an automated pipeline to invert high-resolution ionospheric phase delays from NISAR LA/LB dual-frequency InSAR data, convert GNSS TEC differences into L-band ionospheric phase, and align both products in radar/geographic coordinates.
- Next step: train a super-resolution model using GNSS ionospheric phase as the low-resolution background and NISAR-derived phase fields as high-resolution supervision.

Honors and Awards

2025 **Zhao Jiuzhang Elite Program Scholarship**, School of Earth and Space Sciences
2024 **Rose Fund Endeavor Scholarship**, School of the Gifted Young
2024 **Zhao Jiuzhang Elite Program Scholarship**, School of Earth and Space Sciences
2023 **Freshman Scholarship**, University of Science and Technology of China (USTC)

Experience and Leadership

2025.08 **IPGP Geophysical Field Camp**: conducted geophysical surveys and joined academic seminars with faculty from the Institut de Physique du Globe de Paris (IPGP).
2025.06 – 2026.01 **Head (Minister)**, Dept. of Cultural & Sports in College Student Union